

Hawkeye EQMS · Audit-Only Module (buyer-driven supplier audit, 24-step flow) · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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1. Module overview

Module	Audit-Only Module (buyer-driven supplier audit, 24-step flow)
Pharma purpose	End-to-end buyer-driven supplier audit. Buyer onboards a candidate supplier, runs a pre-qualification, optionally posts an RFQ to engage a 3rd-party auditor, schedules the audit, executes (onsite/remote), generates a report, drives CAPA closure, and monitors the supplier post-audit until the next requalification audit. Spans 24 super-user-defined process steps mapped onto an 8-phase audit lifecycle.
Compliance basis	ICH Q7 §17 (audit of API mfrs) · ICH Q9 (risk-based supplier qualification) · ICH Q10 §2.7 (supplier mgmt) · 21 CFR 211.84 (component qualification) · EU GMP Chapter 7 (outsourced activities) · ISO 19011 (audit programmes) · ISO 13485 §7.4 (purchasing) · WHO TRS 957 Annex 3
Hawkeye routes	/buyer/marketplace, /supplier-marketplace, /rfqs, /rfqs/new, /request-audit, /audits, /audits/[id], /buyer/suppliers/[id]/quality-events (frontend) · /api/audits/buyer, /api/audits/auditor, /api/audits/supplier, /api/audits/:id/phases, /api/audits/:id/prep, /api/audits/:id/plan, /api/audits/:id/agenda, /api/scheduling, /api/audits/:id/questions, /api/evidence, /api/v2/findings, /api/audits/:id/report, /api/audits/:id/report/observations/:observationId/capa, /api/capas, /api/v2/capas, /api/audits/:id/close, /api/buyer/marketplace/suppliers, /api/rfqs, /api/auditor-network, /api/monitoring, /api/suppliers/:id/quality-events (backend)
Hawkeye model file	AuditRequestMaster · Assessment · AuditPlan · AuditAgenda · AuditSchedule · PreAuditQuestionnaire · AuditArtifact · AssessmentFinding · AuditReport · Capa (V1) · AssessmentCapa (V2) · MonitoringSignal · SupplierRiskMetrics · AuditorAffiliation · AuditorQualification · SupplierPreQualification · Template + TemplateQuestions · ReportTemplate · ReportInstance · AuditCycleTemplate · WorkflowDefinition · BatchRecord · Equipment
Tenant module flag	modules.AUDITS

The audit-only module spans the full BUYER → SUPPLIER lifecycle: discover → onboard → pre-qualify → engage auditor → schedule → execute → report → CAPA → close → monitor → requalify. It runs on TWO parallel model lineages today: V1 (AuditRequestMaster + Capa + Evidence) and V2 (Assessment + AssessmentFinding + AssessmentCapa + AssessmentEvidence). V2 stores legacyRefs.auditRequestId to bridge both. The 8-phase lifecycle (INITIATED → PREP → SCOPE_AGENDA → SCHEDULING → EXECUTION → REPORTING → FOLLOWUP_CAPA → CLOSURE) implements the core audit. The 24-step super-user process additionally covers pre-audit (steps 1–11) and post-audit (steps 22–24) work that lives in adjacent modules (Pre-Qual, Auditor Network, Monitoring, Requalification scheduler).

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET = parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
#01 Auditee onboarding — Non-Technical Checklist (buyer Purchase team requests onboarding) ICH Q10 §2.7	SupplierPreQualificationModel.checklist[] (PASS/FAIL/N_A items). Missing: dedicated buyer-purchase-team request form; explicit non-technical-vs-technical split.	PARTIAL	
#02 Auditee Due Diligence — Technical / Primary Checklist (Buyer QA) 21 CFR 211.84	SupplierPreQualificationModel.checklist[]; QualificationMethod for desk review. Missing: structured technical-checklist template; checklist-item-level evidence upload.	PARTIAL	
#03 Sample procurement & feasibility / R&D trial ICH Q7 §17	Not modeled — no sample-procurement / R&D-trial entity, no PO/invoice tracking against samples, no trial-result feed into PQ decision.	GAP	
#04 Provisional Approval (PQ process) ICH Q10 §2.7	SupplierPreQualificationModel decision enum (APPROVED / CONDITIONALLY_APPROVED / REJECTED) + auditRequestId link auto-escalates an APPROVED PQ to a full audit request. /decision endpoint live.	MET	
#05 Pre-Audit Communication (formal intimation) ISO 19011 §6.2	INTIMATION_LETTER artifact type defined in PREP-phase artifact constants; auditor assignment via POST /api/audits/:id/assign-auditors. Missing: structured intimation-letter template; read-receipt tracking.	PARTIAL	
#06 3rd-party auditor due diligence — COI verification + acceptance by Auditee ISO 19011 §7.2 (auditor competence)	AuditorAffiliationModel (PENDING / ACTIVE / REVOKED · INTERNAL / EXTERNAL); AuditorQualificationModel.coiDeclarations[] (declaredAt, hasConflict, conflictDetails); AuditRequestMaster.coiDeclarationSignedAt/By for per-audit sign-off.	MET	
#07 Audit Scheduling (multi-party slot agreement) ISO 19011 §6.3	AuditScheduleModel (status: DRAFT → PROPOSED → HELD → CONFIRMED · mode: REMOTE / ONSITE / HYBRID · multi-party constraints + timezone); /api/scheduling/:id/confirm-dates endpoint.	MET	
#08 Pre-Audit Documentation request (SMF, SOP list, Spec/STP, Stability, process flow) 21 CFR 211.84 · ICH Q7 §17.4	AuditArtifact model with artifactType (SITE_MASTER_FILE, DRL, SOP_LIST, etc.); PREP-phase artifact defaults; POST /api/audits/:id/artifacts. Missing: structured document-request template; per-document checklist completion status.	PARTIAL	
#09 Pre-Audit Questionnaire (auditor designs focus) ISO 19011 §6.3	PreAuditQuestionnaireModel (status: DRAFT → SENT → IN_PROGRESS → SUBMITTED → REVIEWED · responses[]); /api/questionnaires CRUD; AI pre-fill via POST /api/ai-prefill/:auditId.	MET	
#10 Audit Agenda (block-by-block plan) ISO 19011 §6.4	AuditAgendaModel (agendaBlocks[] with startAt/endAt/topic/location · attendees[] · multi-version · status: DRAFT → PROPOSED → CONFIRMED); /api/audits/:id/agenda CRUD.	MET	
#11 Audit Checklist — standard verification (aide-de-memoir) ICH Q7 / WHO TRS 957	AuditQuestions model + GMP_CHECKLIST artifact type. Missing: dedicated aide-de-memoir checklist model with reusable question bank; observation mapping during execution.	PARTIAL	
#12 Audit Execution (onsite / offsite / hybrid) ISO 19011 §6.5	Phase-state machine tracks EXECUTION start/end; RemoteSession model supports virtual audits; /api/audits/:id/questions/:qId/response + /api/evidence + /api/audits/:id/questions/:qId/flag. Missing: real-time co-auditor collaboration; live deficiency-flagging UI on execution screen.	PARTIAL	
#13 Opening Meeting (client + auditee company presentation) ISO 19011 §6.4.2	AuditAgendaModel.agendaBlocks[] schedules opening; OPENING_MEETING_MINUTES artifact type defined. Missing: opening-meeting form template; minutes-capture form; attendee sign-in.	PARTIAL	
#14 Closing Meeting reporting ISO 19011 §6.6	CLOSING_MEETING_MINUTES artifact type defined under FINDINGS phase. Missing: closing-meeting form; preliminary-findings summary; auditee acknowledgment capture.	PARTIAL	
#15 Facility Certification / Communication (Approved pending CAPA / Conditional / Rejected) ICH Q10 §2.7	AuditRequestMaster.facilityOutcome (SATISFACTORY / CONDITIONALLY_SATISFACTORY / UNSATISFACTORY) + facilityOutcomeSetAt/By; CLOSURE-phase artifact set.	MET	

Pharma expectation	Hawkeye implementation	Status	Note
#16 Deficiency Reporting (within 1 week of audit) 21 CFR 211.84 · ICH Q7 §17	AssessmentFindingModel (severity / domain / status · linkedEvidencelds); AuditReport.observations[] with severity + gmpClassification + capaResponseDeadlineDays; CAPA auto-source-ref from findings. Missing: 7-day SLA enforcement; preliminary-vs-final state.	PARTIAL	
#17 Deficiency Validation / Acceptance by Supplier ICH Q10 §2.7	AuditRequestMaster.deficiencyValidation enum (PENDING / ACCEPTED / PARTIALLY_ACCEPTED / DISPUTED) + deficiencyValidationAt/By/Reason; CAPA-V2 triage decisions. Missing: line-item-level dispute UI; dispute-resolution workflow.	PARTIAL	
#18 Audit Report (within 30 days of audit) ICH Q7 §17.7 · ISO 19011 §6.6	AuditReportModel (status: DRAFT → PENDING_REVIEW → APPROVED → PENDING_SIGNATURES → COMPLETED · factualAccuracyReview by supplier · reportApproval by QA · signatures[]); /api/audits/:id/report; AI report assembly via POST /api/ai/audit-agents/assemble-report. 30-day SLA not enforced at model level.	MET	
#19 CAPA Plan (supplier within 30 days of report) 21 CFR 820.100 · ICH Q10 §3.2.2	Capa (V1) + AssessmentCapa (V2) — 7-state machine (DRAFT → NEEDS_SUPPLIER → IN_REVIEW → REWORK_REQUESTED → APPROVED → CLOSED · OVERDUE) · targetDate · actions[] with visibility · auto-source-ref from findings. Missing: 30-day SLA cron.	MET	
#20 Review & Acceptance / Rejection of CAPA Plan 21 CFR 820.100	CAPA REWORK_REQUESTED / APPROVED transitions; CAPA-V2 approval stages + triage decisions. Missing: formal e-sig sign-off on CAPA approval; approval-authority verification.	PARTIAL	
#21 Audit Closure Certification ICH Q7 §17.7	CLOSURE-phase artifact list (FINAL_REPORT, AUDIT_CLOSURE_CERTIFICATE); facilityOutcome field on AuditRequestMaster; AuditReport.status=COMPLETED. Missing: automated closure trigger when all CAPAs CLOSED; closure-form template.	PARTIAL	
#22 CAPA tracking / Follow-up (per target date) 21 CFR 820.100(b)	Capa.actions[] + lastActivityAt + linkedCapalds rollout. OVERDUE status enum exists but no scheduler flips records to OVERDUE today; the cross-module overdue-scan handler covers training / MRM action items / CAPA-V2 action items / equipment but not parent CAPA records by targetDate.	PARTIAL	
#23 Supplier monitoring / rating (monthly) ICH Q10 §2.7 · ISO 9001 §8.4	MonitoringSignal model (severity / status · OPEN / ACKED / RESOLVED · per-tenant/audit/site/product); SupplierRiskMetrics + SupplierRiskSnapshot models; /api/monitoring routes. Missing: monthly cron to compute supplier scorecard; rating-calculation logic; dashboard widget.	PARTIAL	
#24 Follow-up / Qualification Audit Planning (yearly) ICH Q10 §2.7 · ICH Q7 §17.7	QualificationCase.requalDueDate field exists; SURVEILLANCE phase keys present in auditPhases.js with empty artifact list. No cron, no auto-create-new-AuditRequestMaster, no requalification calendar / alert.	GAP	
Buyer Marketplace journey — browse suppliers → view product/API library → start RFQ from listing —	Backend marketplace catalog scaffolded (Postgres) + buyer marketplace API (GET /api/buyer/marketplace/suppliers); /supplier-marketplace page is supplier-facing (wrong audience for buyer browse); /rfqs/new step 0 forces a manual supplier dropdown — no marketplace-driven discovery, no API library page for buyers, no public supplier-profile/product-catalog page.	GAP	
Every quality event names its supplier (Deviation · CAPA · Complaint · ChangeControl · BatchRecord · Equipment) ICH Q10 §2.7 · 21 CFR 211.84	supplierId field on Deviation + Complaint + Capa + AssessmentCapa; ChangeControl.supplierId + auto-flag triggersRequalification when changeType=SUPPLIER; BatchRecord.primarySupplierId + per-BOM-line supplierId; Equipment.vendorSupplierId. All indexed for hot-path queries.	MET	
Unified Supplier Quality Events pane (one screen, all open EQMS work) ICH Q10 §2.7	GET /api/suppliers/:id/quality-events returns 7-category rollup (CAPAs · deviations · complaints · changes · audits · batches · equipment) via SupplierQualityEventAggregator service. Frontend page at /buyer/suppliers/[id]/quality-events with KPI tiles + 7 tabs + SupplierContextBadge.	MET	
Closure-loop automation — regulatory complaint → for-cause supplier audit auto-triggered; CAPA closure → supplier scorecard recompute 21 CFR 820.100 · ICH Q10 §2.7	complaintRoutes hook: requiresRegulatoryReporting + supplierId → triggerForCauseAudit (dedupes per supplier, back-links Complaint.linkedAuditId). capaController hook: status=APPROVED/CLOSED + supplierId → calculateSupplierScorecard + persist SupplierRiskSnapshot.	MET	

Pharma expectation	Hawkeye implementation	Status	Note
AI agents see supplier history when triaging quality events —	complaintTriageService + deviationFiveWhyScaffolder both call buildSupplierContextForAi(supplierId) which injects open-event counts + recently-closed counts + top open items into the prompt. Repeat-offender suppliers up-weight severity; RCA spots systemic patterns.	MET	
V1 audit observation → CAPA per-observation auto-create (parity with V2) 21 CFR 820.100	POST /api/audits/:auditId/report/observations/:observationId/capa — sibling to existing bulk handler. Idempotent (returns reused: true if a CAPA is already linked). Maps severity/target-date from observation classification.	MET	
Module bundle dependency resolution (SUPPLIER_QUALITY → CAPA + EVENT + AUDIT) —	applyModuleBundles() in universalModuleConfigService — SUPPLIER_QUALITY transitively pulls CAPA_MANAGEMENT + EVENT_MANAGEMENT + AUDIT_MANAGEMENT (which itself pulls DOCUMENT_CONTROL). Read-time resolution; saved DB state untouched. promotedBy map tells the UI which modules are force-on by bundle.	MET	
Audit templates seeded for end-to-end demo (Template + TemplateQuestions + ReportTemplate + AuditCycleTemplate + WorkflowDefinition) ICH Q7 §17	scripts/seed-audit-only-users.mjs now creates: Template id=1 (ICH Q7 / 21 CFR 211 Pre-Audit Questionnaire) + 12 TemplateQuestions across 3 categories · ReportTemplate (Standard Pharma Audit Report, 8 blocks) · AuditCycleTemplate (cGMP, 6 phases / 13 milestones) · WorkflowDefinition (AUDIT_MANAGEMENT, 7 phases with required artifacts + role gating).	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Karan Mehta · Buyer Purchase / SCM buyer · purchase	buyer.purchase@acme-pharma.demo	Initiates onboarding (#01), procures samples (#03), drives early commercial engagement.	#01 onboarding #03 samples
Priya Nair · Audit Program Mgr buyer · QA program manager	audit.program@acme-pharma.demo	Owns the buyer-side audit programme: due diligence (#02, #06), pre-audit comm (#05, #08), scheduling (#07), reviews CAPA (#20), drives closure (#21), monitors (#23), schedules requal (#24).	#02 #05–#08 #15 #20–#24 INITIATED CLOSURE
Dr Elena Vasquez · VP Quality (tenant_admin) tenant_admin	vp.quality@acme-pharma.demo	Final approval on facility certification (#15) for HIGH-risk suppliers; signs off requalification programme (#24).	#15 #21 #24
Maria Santos · Lead Auditor (3rd-party · AuditCorp) auditor · lead	audit.lead@auditcorp.demo	Signs COI (#06), designs questionnaire & agenda (#09, #10), executes audit (#12), reports findings (#13–#16), drafts audit report (#18), reviews CAPA (#20), certifies closure (#21).	#06 #09–#21 PLANNING SCHEDULING EXECUTION REPORTING CLOSURE
Rahul Kapoor · Co-Auditor / Reviewer auditor · co-auditor / reviewer	auditor.co@auditcorp.demo	Supports lead in execution (#12), reviews report drafts (#18).	#12 #18
Asha Sharma · Supplier QA Head (Global Pharma) supplier · QA contact	qa.head@globalpharma.demo	Receives intimation, fills pre-audit questionnaire (#09), uploads SMF/SOPs/Spec (#08), hosts audit (#12–#14), drafts CAPA plan (#19), reports CAPA progress (#22).	#01 (response) #08 #09 #12–#14 #17 #19 #22 PREP EXECUTION FOLLOWUP_CAPA

4. Feature catalogue (click-by-click)

4.1 Phase 1 · INITIATED — buyer creates audit request

What it does	Buyer creates an audit request, picks supplier + site + assessment type, assigns auditor(s).
Menu / route	/request-audit, /audits, /buyer/marketplace
Available to roles	buyer buyer-admin
Backend API	POST /api/audits/buyer · POST /api/audits/:id/assign-auditors

CLICK-BY-CLICK

- 1. **NAVIGATE** Buyer opens '/request-audit' (or starts from /buyer/marketplace)
→ Form renders
- 2. **TYPE** Pick supplier (today: dropdown · ideal: from marketplace)
→ Supplier resolved
- 3. **TYPE** Pick site + assessment type
- 4. **CLICK** Submit
→ AuditRequestMaster created · trackStatus='Request Received' · phaseState.INITIATED.status=IN_PROGRESS
- 5. **CLICK** Open audit detail · 'Assign auditors'
→ auditor_id set · auditorDecision=PENDING

4.2 Phase 2 · PREP — supplier completes pre-audit questionnaire + DRL

What it does	Supplier fills the pre-audit questionnaire and uploads the document-requirements list (SMF, SOPs, Spec/STP, Stability, process flow). AI pre-fill optional.
Menu / route	/audits/[id] · supplier-side questionnaire view
Available to roles	supplier auditor
Backend API	POST /api/audits/:id/prep/start · POST /api/questionnaires/:id/submit · POST /api/audits/:id/artifacts/:id/submit
AI assistance	AI POST /api/ai-prefill/auditId — Gemini pre-fills questionnaire from supplier KB.

CLICK-BY-CLICK

- 1. **CLICK** Auditor sends pre-audit questionnaire
→ AuditArtifact[PRE_AUDIT_QUESTIONNAIRE].status=sent · questionnaireStatus=sent_to_supplier
- 2. **TYPE** Supplier fills questionnaire (or runs AI pre-fill, then reviews)
→ Responses saved
- 3. **CLICK** Supplier submits
→ questionnaireStatus=supplier_submitted
- 4. **CLICK** Supplier uploads SMF / SOP list / Spec / STP / Stability
→ AuditArtifact[DRL].status=complete

4.3 Phase 3 · PLANNING (SCOPE_AGENDA) — auditor defines scope + agenda

What it does	Auditor defines audit scope, objectives, risk summary; builds agenda blocks; routes for confirmation.
Menu / route	/audits/[id] · plan + agenda tabs
Available to roles	auditor
Backend API	POST /api/audits/:id/plan · POST /api/audits/:id/plan/submit · POST /api/audits/:id/agenda · POST /api/audits/:id/agenda/confirm

CLICK-BY-CLICK

- 1. **TYPE** Fill scope, objectives, risk summary
→ AuditPlan.status=DRAFT
- 2. **CLICK** Submit plan for approval
→ AuditPlan.status=SUBMITTED
- 3. **TYPE** Build agenda blocks (start/end/topic/location)
→ AuditAgenda.status=DRAFT
- 4. **CLICK** Propose to auditee + confirm
→ AuditAgenda.status=CONFIRMED · milestone AGENDA_FINALIZED=DONE

4.4 Phase 4 · SCHEDULING — confirm dates + logistics

What it does	Confirm audit dates with all parties; capture availability; lock the slot.
Menu / route	/audits/[id] · schedule tab
Available to roles	auditor buyer supplier
Backend API	POST /api/scheduling/:auditId · POST /api/scheduling/:auditId/confirm-dates

CLICK-BY-CLICK

- 1. **TYPE** Pick scheduledDate + mode (REMOTE/ONSITE/HYBRID)
→ AuditSchedule.status=PROPOSED

2. **CLICK** All parties confirm
→ AuditSchedule.status=CONFIRMED · milestone DATES_CONFIRMED=DONE

4.5 Phase 5 · EXECUTION — auditor conducts audit, collects evidence

What it does	Auditor executes audit (onsite/remote), responds to questions, uploads evidence, flags follow-ups.
Menu / route	/audits/[id] · execution tab
Available to roles	auditor
Backend API	POST /api/audits/:id/questions/:qId/response · POST /api/audits/:id/questions/:qId/flag · POST /api/evidence · POST /api/v2/evidence
AI assistance	AI POST /api/ai/wave2/draft-observation — observation suggester from cross-company audits.

CLICK-BY-CLICK

1. **CLICK** Open execution tab · transition to EXECUTION
→ phaseState.EXECUTION=IN_PROGRESS
2. **TYPE** Respond to each audit question · attach evidence files
→ AuditQuestions.responses + Evidence rows
3. **CLICK** Flag question for follow-up (if needed)
→ flagStatus set
4. **CLICK** Hold closing meeting · upload CLOSING_MEETING_MINUTES
→ Milestone CLOSING_MEETING=DONE · trackStatus='Audit Completed'

4.6 Phase 6 · REPORTING (FINDINGS) — auditor drafts report

What it does	Auditor records findings (severity + domain + linkedEvidenceIds); generates draft report; supplier does factual-accuracy review; QA approves; signatures collected.
Menu / route	/audits/[id] · findings + report tabs
Available to roles	auditor supplier (factual review) buyer-QA (approval)
Backend API	POST /api/v2/findings · POST /api/audits/:id/report · POST /api/report-instances
AI assistance	AI POST /api/ai/audit-agents/assemble-report — Gemini drafts narrative report from findings + evidence + ICH Q7 / WHO-GMP context.

CLICK-BY-CLICK

1. **TYPE** Create finding (severity, domain, description, linked evidence)
→ AssessmentFinding.status=OPEN
2. **CLICK** Run AI report assembler
→ AuditReport.status=DRAFT with rendered narrative
3. **CLICK** Send for supplier factual-accuracy review
→ AuditReport.status=PENDING_REVIEW
4. **CLICK** QA approves report
→ AuditReport.status=APPROVED
5. **CLICK** Collect signatures
→ AuditReport.status=PENDING_SIGNATURES → COMPLETED · milestone FINAL_REPORT=DONE

4.7 Phase 7 · CAPA (FOLLOWUP_CAPA) — supplier addresses findings

What it does	Each finding spawns a CAPA. Supplier drafts plan, submits for review, auditor approves or requests rework. CAPAs reach APPROVED → CLOSED.
Menu / route	/audits/[id] · capa tab · /capa-v2
Available to roles	supplier (plan) auditor (review) buyer (close-out)
Backend API	POST /api/capas · PATCH /api/capas/:id/status · POST /api/v2/capas · PATCH /api/v2/capas/:id/status
AI assistance	AI POST /api/ai/capa/rca-draft — RCA drafter (5-Whys / fishbone) helps supplier author plan.

CLICK-BY-CLICK

1. **CLICK** Auditor creates CAPA from finding
→ Capa.status=DRAFT → NEEDS_SUPPLIER
2. **TYPE** Supplier drafts plan (RCA, action items, target dates)
→ actions[] populated

3. **CLICK** Supplier submits plan
→ Capa.status=IN_REVIEW
4. **CLICK** Auditor approves (or requests rework)
→ Capa.status=APPROVED (or REWORK_REQUESTED → IN_REVIEW)
5. **CLICK** Buyer verifies completion
→ Capa.status=CLOSED

4.8 Phase 8 · CLOSURE — buyer signs off, audit closed

What it does	Buyer reviews everything, assigns final compliance classification (NAI / VAI / OAI), signs report, closes audit.
Menu / route	/audits/[id] · closure tab
Available to roles	buyer
Backend API	POST /api/audits/:id/report/sign · POST /api/audits/:id/close

CLICK-BY-CLICK

1. **CLICK** Sign report
→ signatures[] appended
2. **CLICK** Set facility outcome (SATISFACTORY / CONDITIONAL / UNSAT)
→ facilityOutcome set
3. **CLICK** Close audit
→ trackStatus='Audit Closed' · all phaseState COMPLETED · AUDIT_CLOSURE_CERTIFICATE artifact

4.9 Audit templates — Template + TemplateQuestions + ReportTemplate + AuditCycleTemplate + WorkflowDefinition

What it does	Five template tiers that back every audit: question bank (Template + TemplateQuestions) · report layout (ReportTemplate, blocks rendered into ReportInstance per audit) · phase + milestone definition (AuditCycleTemplate, per tenant + module) · workflow engine config (WorkflowDefinition with required artifacts per phase + role gating).
Menu / route	Seeded by `node scripts/seed-audit-only-users.mjs`
Available to roles	tenant_admin
Backend API	GET /api/templates · GET /api/templates/:id/questions · GET /api/report-templates · GET /api/workflow-definitions

CLICK-BY-CLICK

1. **WAIT** Run seed
→ 1 Template (id=1, ICH Q7 / 21 CFR 211 Pre-Audit Questionnaire) + 12 TemplateQuestions (3 categories × 4 each) + 1 ReportTemplate (8 blocks) + 1 AuditCycleTemplate (cGMP, 6 phases / 13 milestones) + 1 WorkflowDefinition (AUDIT_MANAGEMENT, 7 phases)
2. **NAVIGATE** PreAuditQuestionnaire seeded for the PREP audit references this template — supplier opens it and sees the 12 questions
→ Questionnaire renders
3. **NAVIGATE** Audit report generation reads ReportTemplate.blocks + populates from audit observations + CAPAs + facility outcome
→ ReportInstance created

4.10 Unified Supplier Quality Events page (Tier-2 EQMS←Supplier bridge)

What it does	One screen that shows all open EQMS work for one supplier — CAPAs · deviations · complaints · changes · active audits · open batches · vendor equipment. Drives the buyer's renewal + risk decisions.
Menu / route	/buyer/suppliers/[id]/quality-events
Available to roles	buyer · buyer_admin · qa_head · vp_quality
Backend API	GET /api/suppliers/:id/quality-events

CLICK-BY-CLICK

1. **NAVIGATE** Click a supplier from /buyer/suppliers
→ Land on supplier detail with Risk + Quality Events tabs
2. **CLICK** Open Quality Events tab
→ SupplierContextBadge + 8 KPI tiles + 7 sub-tabs
3. **CLICK** Click any tab (CAPAs, Deviations, Complaints, Changes, Audits, Batches, Equipment)
→ Tab loads list + severity chips
4. **CLICK** Toggle 'Show recently-closed' switch
→ Footer adds 90-day closed counts

4.11 Closure-loop automation (complaint → for-cause audit · CAPA closure → scorecard refresh)

What it does	Two server-side hooks: a regulatory complaint with a supplier link auto-triggers a for-cause supplier audit (deduped per supplier); CAPA closure on a supplier-linked CAPA recomputes the supplier scorecard and writes a new SupplierRiskSnapshot.
Menu / route	(no UI) – backend hooks fire on POST/PUT /api/complaints + PATCH /api/capas/:id/status
Available to roles	(automatic)
Backend API	POST /api/complaints (hook) · PATCH /api/capas/:id/status (hook)

CLICK-BY-CLICK

- WAIT** Buyer creates a critical safety complaint with supplierId set
→ complaint created · requiresRegulatoryReporting auto=true · MDR clock set
- WAIT** Server hook fires triggerForCauseAudit
→ New AuditRequestMaster created · auditType=FOR_CAUSE · complaint.linkedAuditId back-filled
- WAIT** Re-running on same complaint
→ Dedupe — second call returns created=false, existingId=...
- WAIT** Auditor approves a CAPA on the supplier
→ SupplierRiskSnapshot row written · band recomputed within seconds

4.12 Buyer Marketplace journey (browse → profile → start RFQ)

What it does	Buyer browses an aggregated marketplace of suppliers, opens a supplier profile (sites + product catalog + audit history + risk), starts an RFQ or audit-request from the listing.
Menu / route	/buyer/marketplace, /supplier-marketplace, /rfqs/new
Available to roles	buyer
Backend API	GET /api/buyer/marketplace/suppliers · GET /api/buyer/marketplace/suppliers/:id/sites · POST /api/rfqs

CLICK-BY-CLICK

- NAVIGATE** (today) Buyer goes straight to /rfqs/new and types supplier name
→ Manual dropdown — no marketplace browse
- NAVIGATE** (needed) Buyer browses /buyer/marketplace · filters by API/category
→ Listing of onboarded + public suppliers
- CLICK** (needed) Opens supplier profile · sees product catalog + audit history
→ Profile + catalog drawer
- CLICK** (needed) Click 'Start RFQ' from supplier profile
→ RFQ pre-filled with supplier + site

4.13 Auditor Network — engage 3rd-party auditor (COI verification + acceptance)

What it does	Buyer browses auditor pool, invites auditor, auditor signs COI declaration, buyer accepts, audit links to auditor org.
Menu / route	/auditor-network · /audits/[id] · auditor tab
Available to roles	buyer auditor
Backend API	GET /api/auditor-network · POST /api/audits/:id/assign-auditors · POST /api/auditor/coi-declaration

CLICK-BY-CLICK

- CLICK** Buyer browses auditor pool
→ AuditorAffiliation[ACTIVE] list
- CLICK** Buyer invites auditor
→ AuditorAffiliation.status=PENDING for that audit
- TYPE** Auditor signs COI declaration
→ AuditorQualification.coiDeclarations[] entry · coiDeclarationSignedAt set
- CLICK** Buyer accepts auditor
→ AuditRequestMaster.auditor_id set · auditorDecision=ACCEPTED

4.14 AI · Pre-Audit Questionnaire pre-fill (Gemini)

What it does	AI pre-fills the supplier's questionnaire from supplier KB + ICH Q7 context. Supplier reviews + edits + submits.
Menu / route	/audits/[id] · prep tab · 'AI pre-fill' button
Available to roles	supplier

Backend API	POST /api/ai-prefill/:auditId
AI assistance	AI Gemini Flash-Lite · structured-output · KB-grounded.

CLICK-BY-CLICK

1. **CLICK** Click 'AI pre-fill'
→ Suggested answers populated · supplier reviews each one
2. **CLICK** Supplier accepts / edits / submits
→ PreAuditQuestionnaire.status=SUBMITTED

4.15 AI · Supplier-Intel agent (public-data fusion)

What it does	Pull openFDA + FDA warning letters + import alerts + EMA EudraGMDP + WHO PQ for the supplier; verdict (known_tenant / public_only / ambiguous / unknown).
Menu / route	/audits/[id] · 'Check public signals' · /supplier-prequalification (also)
Available to roles	any tenant user
Backend API	POST /api/ai/audit-agents/supplier-intel
AI assistance	AI Public-data fusion · LLM for narrative.

CLICK-BY-CLICK

1. **CLICK** Click 'Check public signals'
→ Drawer opens
2. **WAIT** Wait ~3-5s for public-data fusion
→ openFDA + warning letters + import alerts + verdict shown

4.16 AI · Audit Report assembler

What it does	Gemini drafts the audit report narrative from findings + evidence + ICH Q7 / WHO-GMP standards.
Menu / route	/audits/[id] · report tab · 'AI assemble report' button
Available to roles	auditor
Backend API	POST /api/ai/audit-agents/assemble-report
AI assistance	AI Gemini Flash-Lite · KB-grounded · factual-citation footnotes.

CLICK-BY-CLICK

1. **CLICK** Click 'AI assemble report'
→ Draft narrative populated
2. **CLICK** Auditor reviews + saves
→ AuditReport.status=DRAFT

4.17 AI · Observation drafter (Wave-2)

What it does	During execution, auto-suggest observation wording from similar findings across other audits.
Menu / route	/audits/[id] · execution tab · per-question 'AI suggest' button
Available to roles	auditor
Backend API	POST /api/ai/wave2/draft-observation
AI assistance	AI Cross-tenant retrieval (anonymized) + LLM rewrite.

CLICK-BY-CLICK

1. **CLICK** Click 'AI suggest observation'
→ Suggested wording shown

4.18 AI · CAPA RCA drafter

What it does	Draft a 5-Whys / fishbone RCA for a CAPA from the finding text.
Menu / route	/audits/[id] · capa tab · 'AI draft RCA' button
Available to roles	supplier
Backend API	POST /api/ai/capa/rca-draft

CLICK-BY-CLICK

1. **CLICK** Click 'AI draft RCA'
→ RCA scaffold populated

4.19 Supplier monitoring (post-closure)

What it does	After audit closure, the supplier is monitored for adverse signals (FDA warning letters, recalls, complaints, expiring certs). Signals raised against supplier/audit/site/product.
Menu / route	/buyer/suppliers/[id]/risk · /monitoring
Available to roles	buyer-QA
Backend API	GET /api/monitoring · POST /api/monitoring/ack

CLICK-BY-CLICK

1. **NAVIGATE** Open buyer supplier risk view
→ Risk metrics + open monitoring signals

4.20 Requalification audit planning (yearly)

What it does	When a supplier's qualification approaches its requalDueDate, automatically alert the buyer + scaffold a new surveillance audit request linked to the previous one.
Menu / route	/audits · '(needed) Requalification calendar'
Available to roles	buyer-QA tenant_admin
Backend API	(needed) GET /api/audits/requalification-due · POST /api/audits/buyer (with surveillance template)

CLICK-BY-CLICK

1. **NAVIGATE** (needed) Buyer opens requalification calendar
→ List of suppliers due for re-audit in next 90 days

5. Full lifecycle walkthrough — one transaction, end-to-end

One supplier walked end-to-end from initial onboarding (#01) through audit closure (#21) — covering the 8 phases of the data-flow doc, mapped onto the 24 super-user steps.

#	Persona	State transition	Action	API	Observed	Result
1	Karan Buyer Purchase	— → PQ DRAFT	Initiates onboarding (#01) → creates Supplier Pre-Qual record (initialRiskBand=MEDIUM, scope, regulatoryStandards).	POST /api/supplier-prequalifications	PQ-YYYY-NNNN created · status=DRAFT	PASS
Expected DB state after this step:						
supplier-prequalifications { _id, pqNumber, supplierName, scope, initialRiskBand, status: 'DRAFT', initiatedBy }						
2	Asha Supplier QA	PQ DRAFT → PQ SUBMITTED	Fills technical checklist (#02) + uploads cert documents.	PUT /api/supplier-prequalifications/:id (status=SUBMITTED)	Status flips · checklist[] populated	PASS
3	Priya Buyer QA	PQ SUBMITTED → PQ APPROVED	Reviews + decides (#04 Provisional Approval).	POST /api/supplier-prequalifications/:id/decision	decision=APPROVED · validUntil=+2y · auditRequestId field ready to bind	PASS
4	Priya Buyer QA	(none) → AUDIT INITIATED	Creates audit request from approved PQ (#05 Pre-Audit Communication, #06 COI, #07 Scheduling).	POST /api/audits/buyer · POST /api/audits/:id/assign-auditors	AuditRequestMaster created · phaseState.INITIATED=IN_PROGRESS · INTIMATION_LETTER artifact created · auditor_id set · auditorDecision=PENDING	PASS
Expected DB state after this step:						
auditrequestmasters { _id, supplier_id, auditor_id, site_id, assessmentTypeId, trackStatus:'Request Received', questionnaireStatus:'request_received', auditorDecision:'PENDING', supplierDecision:'PENDING', phaseState:{ INITIATED:{status:'IN_PROGRESS'} } }						

#	Persona	State transition	Action	API	Observed	Result
5	Maria Lead Auditor	AUDIT INITIATED → PREP	Signs COI (#06) · sends pre-audit questionnaire (#08, #09).	POST /api/auditor/coi-declaration · POST /api/audits/:id/prepare/start · POST /api/audits/:id/artifacts (PRE_AUDIT_QUESTIONNAIRE) · POST /api/audits/:id/artifacts/:id/send	AuditorQualification.coiDeclarations[] entry · phaseState.PREP=IN_PROGRESS · questionnaireStatus=sent_to_supplier	PASS
6	Asha Supplier QA	PREP → PREP COMPLETE	Uploads SMF / SOPs / Spec / STP / Stability (#08) · fills questionnaire with optional AI pre-fill (#09) · submits.	POST /api/ai-prefill/:auditId · POST /api/questionnaires/:id/submit · POST /api/audits/:id/artifacts/:id/submit	AuditArtifact[DRL].status=complete · PreAuditQuestionnaire.status=SUBMITTED · questionnaireStatus=supplier_submitted	PASS
7	Maria Lead Auditor	PREP COMPLETE → PLANNING (SCOPE_AGENDA)	Defines scope + objectives + risk summary (#11 Audit Checklist) · builds agenda (#10) · proposes + confirms.	POST /api/audits/:id/plan/submit · POST /api/audits/:id/agenda/confirm	AuditPlan.status=APPROVED · AuditAgenda.status=CONFIRMED · milestone AGENDA_FINALIZED=DONE · phaseState.SCOPE_AGENDA=COMPLETED	PASS
8	Maria Lead Auditor	PLANNING → SCHEDULING DONE	Confirms dates + mode (#07 finalised).	POST /api/scheduling/:auditId/confirm-dates	AuditSchedule.status=CONFIRMED · milestone DATES_CONFIRMED=DONE	PASS
9	Maria + Rahul Auditors	SCHEDULING DONE → EXECUTION	Run audit (#12) · opening meeting (#13) · respond to questions + upload evidence · closing meeting (#14).	POST /api/audits/:id/questions/:qId/response · POST /api/evidence · POST /api/audits/:id/artifacts (CLOSING_MEETING_MINUTES)	AuditQuestions responses filled · Evidence rows · trackStatus='Audit Completed' · milestone CLOSING_MEETING=DONE	PASS
10	Maria Lead Auditor	EXECUTION → REPORTING	Records findings (#16 Deficiency Reporting) · runs AI report assembler · supplier reviews factual accuracy (#17) · QA approves · signatures.	POST /api/v2/findings · POST /api/ai/audit-agents/assemble-report · POST /api/audits/:id/report	AssessmentFinding rows · AuditReport.status=DRAFT → PENDING_REVIEW → APPROVED → COMPLETED · milestone FINAL_REPORT=DONE	PASS
11	Asha Supplier QA	REPORTING → FOLLOWUP_CAPA	Drafts CAPA plan per finding (#19) — uses AI RCA drafter — submits.	POST /api/v2/capas · POST /api/ai/capa/rca-draft · PATCH /api/v2/capas/:id/status (IN_REVIEW)	AssessmentCapa rows · status=IN_REVIEW · actions[] populated	PASS
12	Maria Lead Auditor	CAPA_IN_REVIEW → CAPA APPROVED → CLOSED	Reviews + approves CAPAs (#20) · buyer verifies + closes (#21).	PATCH /api/v2/capas/:id/status (APPROVED → CLOSED)	All AssessmentCapa rows CLOSED	PASS
13	Priya Buyer QA	CAPA_CLOSED → AUDIT_CLOSED	Sets facility outcome (#15) · signs report · closes audit (#21).	POST /api/audits/:id/report/sign · POST /api/audits/:id/close	facilityOutcome=SATISFACTORY · trackStatus='Audit Closed' · all phaseState COMPLETED · AUDIT_CLOSURE_CERTIFICATE artifact	PASS
Expected DB state after this step: <pre>auditrequestmasters { facilityOutcome:'SATISFACTORY', facilityOutcomeSetAt, facilityOutcomeSetBy, trackStatus:'Audit Closed', high_status:5, complianceStatus:'compliant', phaseState:{ all phases:'COMPLETED' } }</pre>						
14	Priya Buyer QA	AUDIT_CLOSED → MONITORING	Supplier moves to monitoring (#23) — risk metrics + adverse-signal feed running.	(automatic) MonitoringSignal subscriptions live · SupplierRiskMetrics snapshots running	MonitoringSignal rows when public-data scanner finds adverse signals	PARTIAL — MONTHLY SCORECARD CRON NOT YET WIRED
15	Priya Buyer QA	MONITORING → REQUALIFICATION DUE (yearly)	(needed) Requalification scheduler fires when QualificationCase.requalDueDate passes; auto-creates surveillance audit (#24).	(needed) cron → POST /api/audits/buyer with surveillance template	—	GAP — NO SCHEDULER TODAY

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Pre-Audit Questionnaire Pre-fill	PREP	POST /api/ai-prefill/:auditId	/audits/[id] · prep tab · 'AI pre-fill'	Gemini pre-fills supplier questionnaire from supplier KB + ICH Q7	Gemini Flash-Lite (free tier)

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Supplier-Intel Agent	INITIATED PREP	POST /api/ai/audit-agents/supplier-intel	/audits/[id] · /supplier-prequalification	Public-data fusion — openFDA + FDA WLs + EU EudraGMDP + WHO PQ + verdict	Public-data fusion + LLM narrative
Audit Report Assembler	REPORTING	POST /api/ai/audit-agents/assemble-report	/audits/[id] · report tab · 'AI assemble report'	Drafts narrative report from findings + evidence + ICH Q7 / WHO-GMP	Gemini Flash-Lite (free tier)
Observation Drafter (Wave-2)	EXECUTION	POST /api/ai/wave2/draft-observation	/audits/[id] · execution tab · per-question 'AI suggest'	Suggests observation wording from cross-company anonymized findings	Cross-tenant retrieval + LLM rewrite
CAPA RCA Drafter	FOLLOWUP_CAPA	POST /api/ai/capa/rca-draft	/audits/[id] · capa tab · 'AI draft RCA'	5-Whys / fishbone scaffold from finding text	Gemini Flash-Lite (free tier)
Autofill Form (generic)	PREP PLANNING	POST /api/ai/audit-agents/autofill-form	auditor + supplier forms	Generic structured-form filler	Gemini Flash-Lite (free tier)

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
PRE_QUAL	ICH Q10 §2.7 21 CFR 211.84	SupplierPreQualification.checklist[] + decision + decidedBy + decisionAt + validUntil	supplier-prequalifications
INITIATED	ISO 19011 §6.2	AuditRequestMaster created · INTIMATION_LETTER artifact · auditor_id assigned	audit-requests-master, audit-artifacts
PREP	ICH Q7 §17.4	PreAuditQuestionnaire SUBMITTED · DRL artifacts complete (SMF, SOPs, Spec/STP, Stability)	pre-audit-questionnaires, audit-artifacts
PLANNING	ISO 19011 §6.4	AuditPlan APPROVED · AuditAgenda CONFIRMED	audit-plans, audit-agendas
SCHEDULING	ISO 19011 §6.3	AuditSchedule CONFIRMED · DATES_CONFIRMED milestone DONE	audit-schedules, workflow-milestone-instances
EXECUTION	ISO 19011 §6.5 ICH Q7 §17	AuditQuestions responses + Evidence + CLOSING_MEETING_MINUTES artifact	audit-questions, evidences, assessment-evidences
REPORTING	ISO 19011 §6.6 ICH Q7 §17.7	AssessmentFinding rows · AuditReport COMPLETED with signatures[]	assessment-findings, audit-reports
FOLLOWUP_CAPA	21 CFR 820.100 ICH Q10 §3.2.2	AssessmentCapa rows reach APPROVED → CLOSED · actions[]	capas, assessment-capas
CLOSURE	ICH Q7 §17.7 ICH Q10 §2.7	facilityOutcome set · AuditReport COMPLETED · AUDIT_CLOSURE_CERTIFICATE artifact	audit-requests-master, audit-reports, audit-artifacts
MONITORING	ICH Q10 §2.7 ISO 9001 §8.4	MonitoringSignal rows · SupplierRiskMetrics snapshots	monitoring-signals, supplier-risk-metrics, supplier-risk-snapshots

8. Test results matrix

Suite	Scope	Result	Evidence
audit-lifecycle.spec.ts	INITIATED → PREP → PLANNING → SCHEDULING → EXECUTION → REPORTING → FOLLOWUP_CAPA → CLOSURE	PASS	8/8 phases · v2 + v1 lineage · audit-test-results.pdf
pre-qual-lifecycle.spec.ts	PQ DRAFT → SUBMITTED → UNDER_REVIEW → APPROVED + auto-link to audit	PASS	4/4 PASS
auditor-coi.spec.ts	Auditor COI declaration + acceptance	PASS	covered
rfq-lifecycle.spec.ts	RFQ DRAFT → AWARDED → CONVERTED to audit-requests-master	PASS	covered
test-supplier-quality-events.mjs (Tier 1)	supplierId persists on Deviation+ChangeControl · aggregator returns rows · AI context summary · module bundles transitively resolve	PASS	20/20 PASS

Suite	Scope	Result	Evidence
test-tier2-supplier-bridge.mjs (Tier 2)	complaint→for-cause audit (deduped + back-linked) · CAPA closure→scorecard recompute + snapshot persisted · aggregator surfaces for-cause audit immediately	PASS	12/12 PASS
test-tier3-supplier-bridge.mjs (Tier 3)	BatchRecord primarySupplierId + BOM-line \$or · Equipment vendorSupplierId + overdue calibration · per-observation V1 audit→CAPA helper (idempotent)	PASS	16/16 PASS
buyer-marketplace.spec.ts	buyer browses suppliers → opens profile → starts RFQ	MISSING	no UI today
requalification-scheduler.spec.ts	yearly auto-trigger of surveillance audit	MISSING	no scheduler today

9. Known gaps + roadmap

- **[SHIPPED v1.2] Tier 1: supplierId on Deviation + ChangeControl + aggregator service + AI agents wired + module bundles** — Closes the schema-foundation gaps. supplierId added to Deviation + ChangeControl with indexes + sourceFromSupplier flag; SupplierQualityEventAggregator service queries 7 collections by supplierId; complaint-triage + deviation-5why agents now inject supplier history into prompts; SUPPLIER_QUALITY module auto-pulls CAPA + EVENT + AUDIT (transitive: AUDIT → DOCUMENT_CONTROL). **DONE**
- **[SHIPPED v1.2] Tier 2: unified Quality Events page + complaint→for-cause hook + CAPA→scorecard hook** — /buyer/suppliers/[id]/quality-events page with KPI tiles + 7 tabs. Hooks: requiresRegulatoryReporting + supplierId → triggerForCauseAudit (deduped, back-linked); CAPA APPROVED/CLOSED + supplierId → calculateSupplierScorecard + persist SupplierRiskSnapshot. **DONE**
- **[SHIPPED v1.2] Tier 2.5: SupplierContextBadge component (reusable across detail pages)** — Drop-in card or inline variant. Wired into the quality-events page header. Ready to slot into Deviation/CAPA/Complaint detail pages when those land. **DONE**
- **[SHIPPED v1.2] Tier 3: BatchRecord primarySupplierId + Equipment vendorSupplierId + V1 audit observation → CAPA** — BatchRecord top-level supplierId + per-BOM-line. Equipment vendorSupplierId + index. Per-observation idempotent endpoint POST /api/audits/:id/report/observations/:observationId/capa. **DONE**
- **[SHIPPED v1.2] Audit templates seeded — Template + 12 TemplateQuestions + ReportTemplate + AuditCycleTemplate + WorkflowDefinition** — scripts/seed-audit-only-users.mjs now creates the full template stack so the demo end-to-end flow actually works (PreAuditQuestionnaire's templateId=1 reference now resolves). **DONE**
- **Buyer Marketplace UI (browse → profile → product/API library → start RFQ)** — Backend marketplace catalog + buyer-marketplace API exist. Build /buyer/marketplace listing, supplier-profile page (sites + product catalog + audit history + risk), API library page, and 'Browse Marketplace' link on /rfqs/new step 0. **HIGH**
- **Requalification / Surveillance auto-scheduler (#24)** — Vercel cron daily → scans QualificationCase.requalDueDate; T-90 alert; T-0 auto-creates surveillance AuditRequestMaster from previous audit + scope template. **HIGH**
- **Supplier scorecard / monthly rating (#23)** — Vercel cron monthly → computes SupplierRiskMetrics snapshot from MonitoringSignal + audit outcomes + CAPA on-time rate; dashboard widget on /buyer/suppliers/[id]. **HIGH**
- **Sample procurement / R&D trial workflow (#03)** — New SampleProcurement model + UI (PO ref, lot, R&D trial result, link back to PQ decision). Today this is paper / email. **HIGH**
- **CAPA target-date OVERDUE scheduler (#22)** — Extend overdueScanService to flip parent Capa / AssessmentCapa records to OVERDUE when targetDate passes; notification-outbox alert; surface on supplier + buyer dashboard. **MEDIUM**
- **Opening + closing meeting forms (#13, #14)** — Structured meeting-minutes form with attendee sign-in + preliminary-findings summary; auditee acknowledgment capture. **MEDIUM**
- **Aide-de-memoir checklist model (#11)** — Reusable GMP checklist library (per ICH Q7 / WHO TRS 957) with question bank · clone-into-audit + observation mapping. **MEDIUM**
- **Per-document checklist completion status (#08)** — Per-DRL-item completion status (sent / received / accepted / rejected) instead of one artifact-level status. **MEDIUM**
- **30-day SLA enforcement on audit report + CAPA plan (#18, #19)** — Cron-flagged OVERDUE state on AuditReport.status if no PENDING_SIGNATURES within 30 d of EXECUTION close; same for CAPA plan submission. **MEDIUM**
- **Line-item deficiency dispute UI (#17)** — Dispute model already exists (deficiencyValidation enum). Add per-finding dispute drawer + resolution workflow. **LOW**
- **Formal e-signature on CAPA approval (#20)** — Wire requireESignature middleware on CAPA APPROVED transition (parallel to deviation closure). **LOW**
- **Centralized StatusTransitionEngine** — Per status-engine-analysis.md: 131+ direct .status= mutations across 20+ controllers. Build centralized engine for validation + audit-trail consistency. **LOW**